Report: Instrumentation Amplifier, Common-Mode Effects, and Filtering

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Part 1: Instrumentation Amplifier with Common-Mode Signal

1. Schematic of Instrumentation Amplifier with Inputs and Common-Mode Source

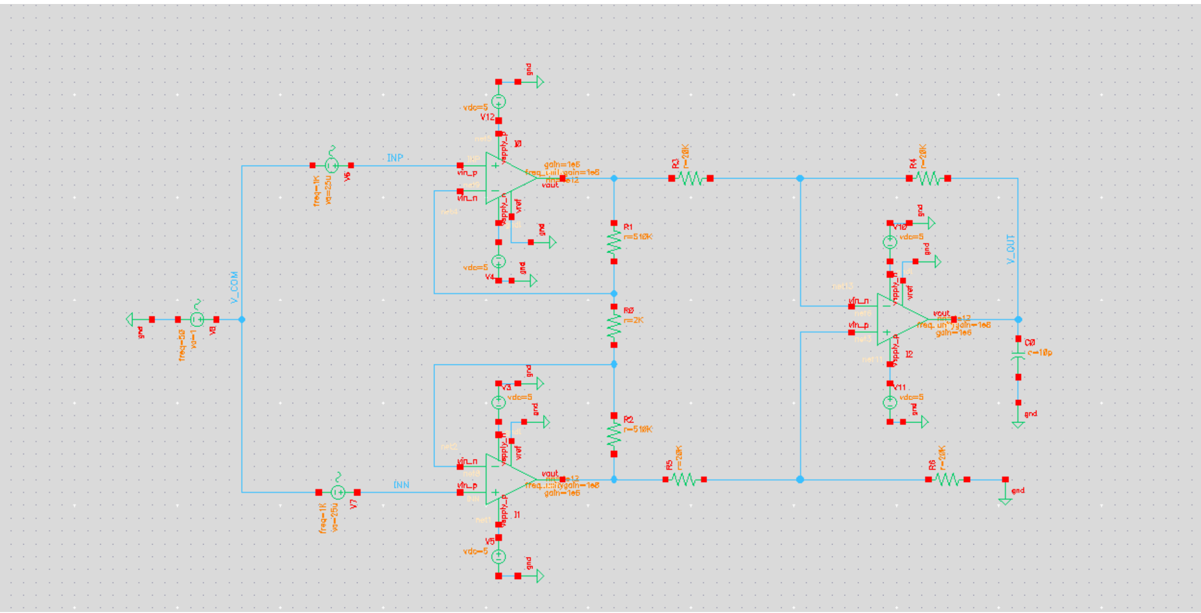
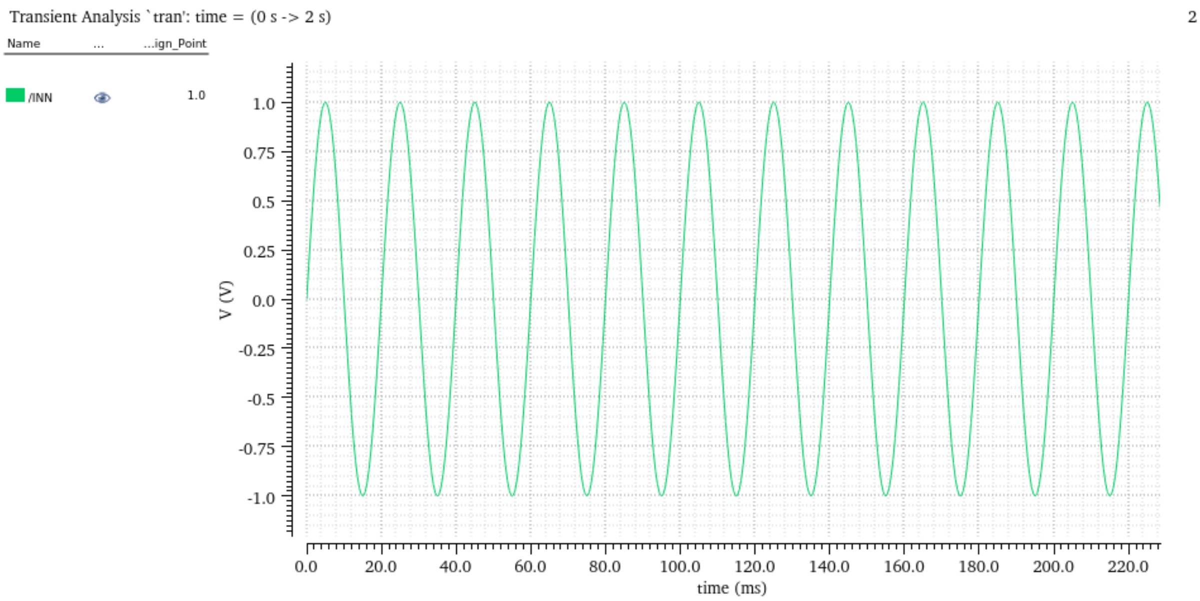


Figure 1.1: Schematic of Instrumentation Amplifier with differential inputs (V1, V2) and common-mode source (V6).

2. Input Signal Plots



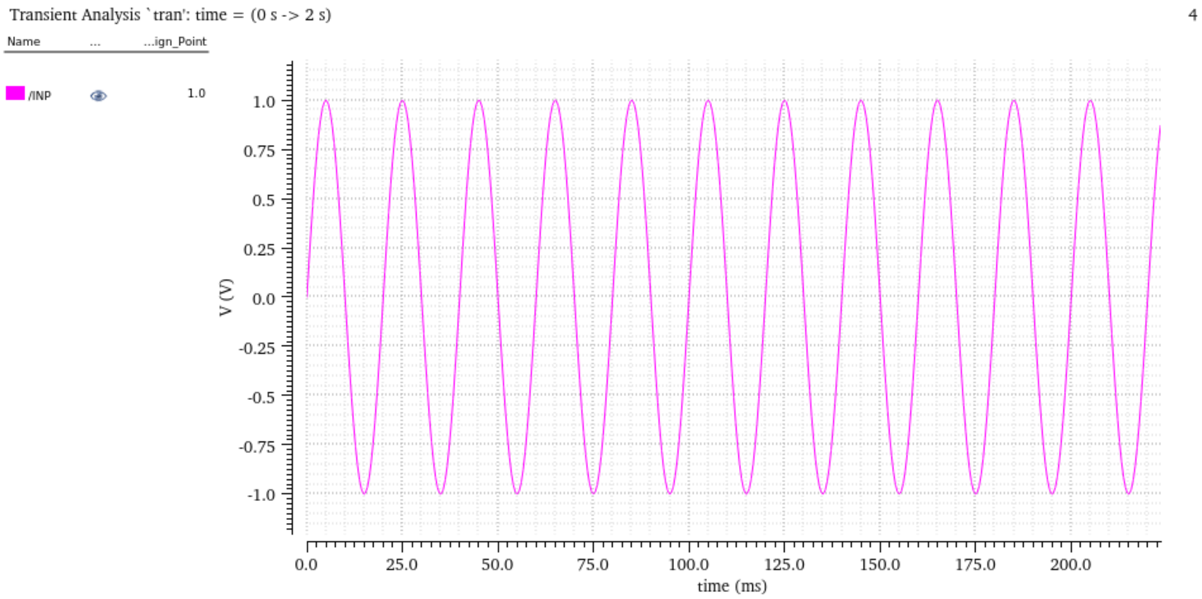


Figure 1.2: Plot of input signals INN and INP (1 kHz, ±0.025 mV, 180° out of phase).

3. Common-Mode Signal Plot

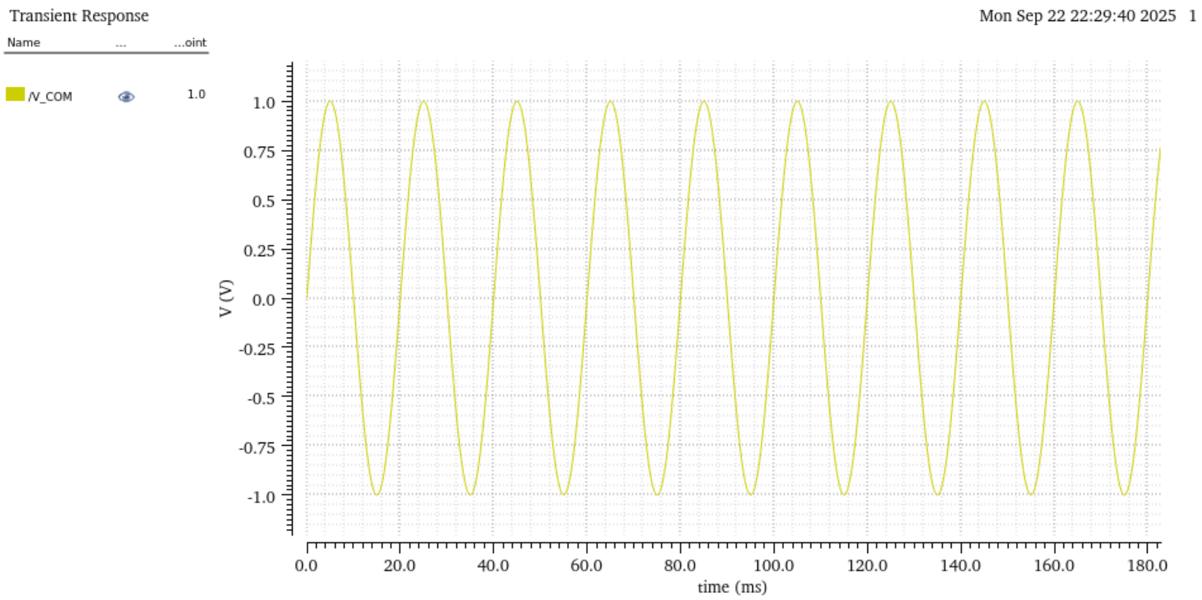


Figure 1.3: Plot of common-mode signal V6 (50 Hz, 1 V amplitude).

4. Output Signal Plot

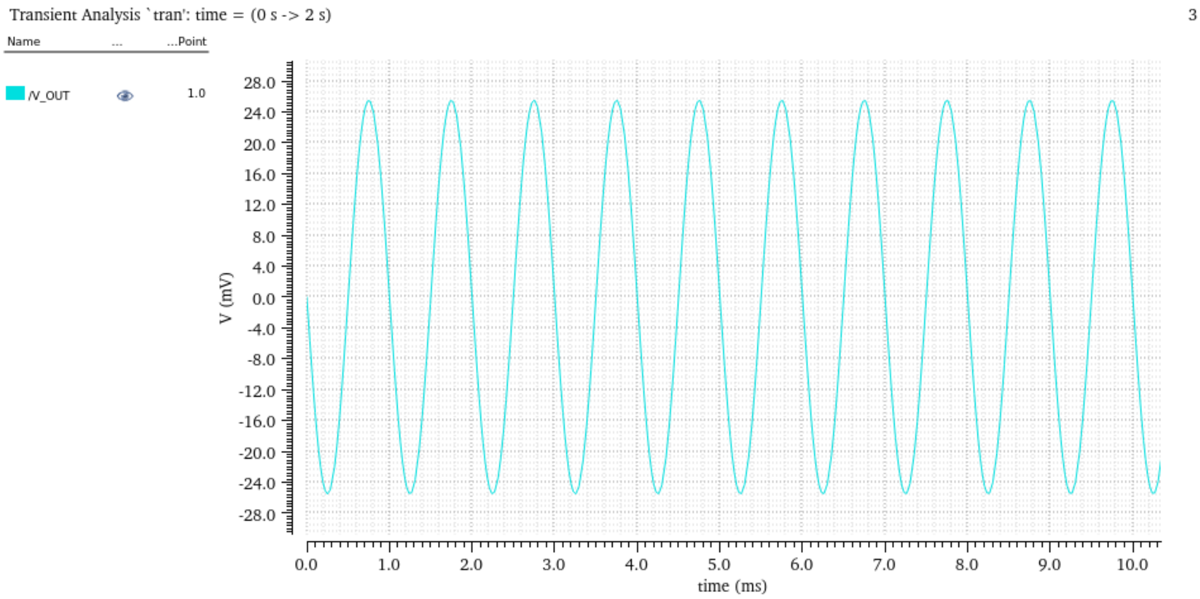


Figure 1.4: Plot of amplifier output showing combined effect of input and common-mode signals.

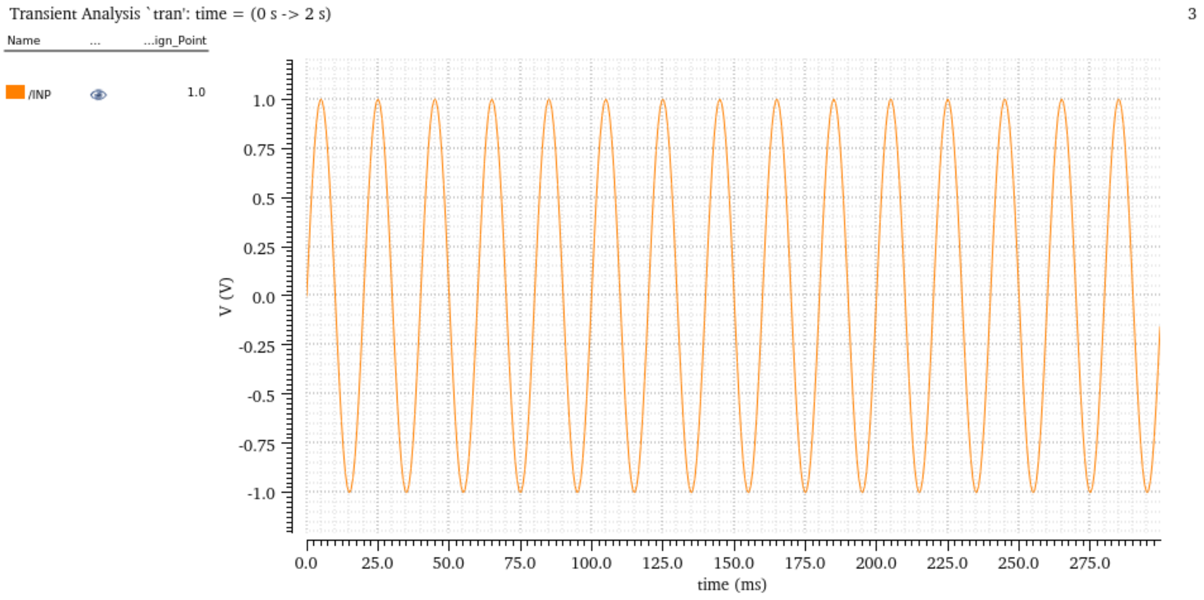
- Peak-to-Peak Value of Output: ______ V
- Simulated Total Gain (at 1 kHz): ______

Part 2: Effect of Resistor Mismatch in IA

1. Schematic with R2 Mismatch

Figure 2.1: Schematic of IA with resistor mismatch (R2 = 2.2 kΩ on non-inverting side).

2. Input Plots (1 kHz Differential Input)



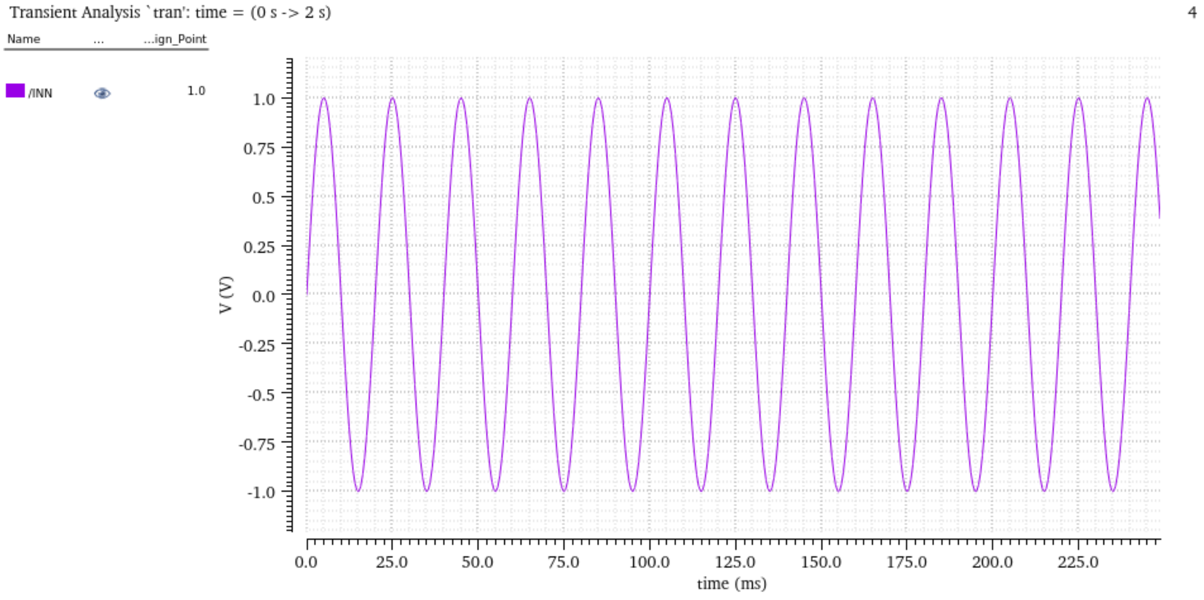


Figure 2.2: Plot of inputs at 1 kHz.

3. Common-Mode (50 Hz)

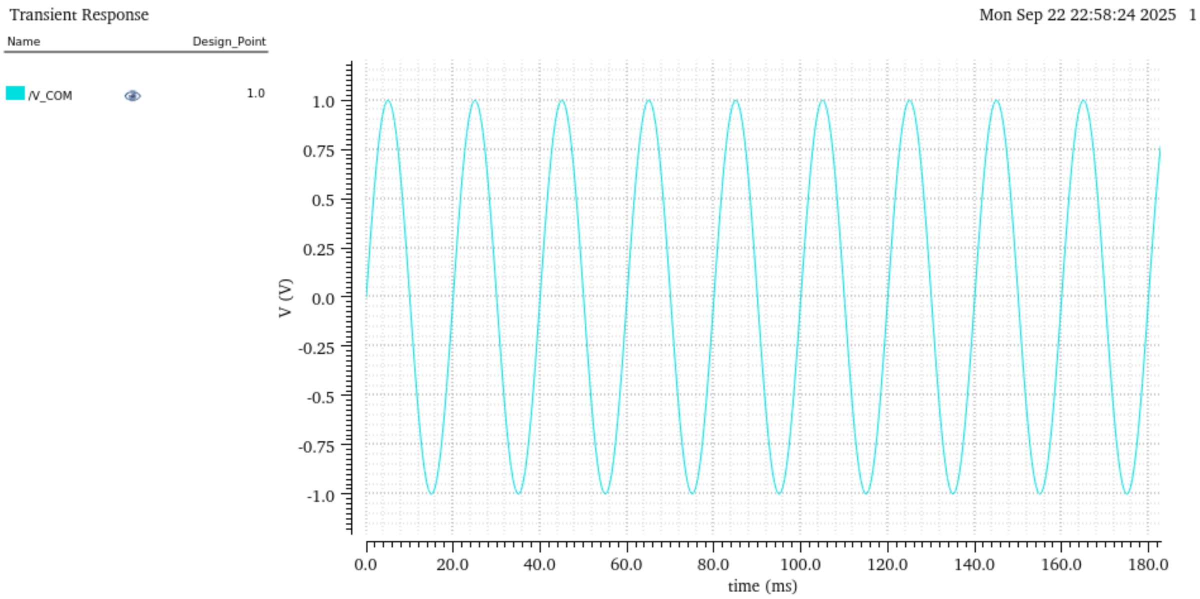


Figure 2.3: Plot of 50 Hz common-mode input

4. Output Signal Plot

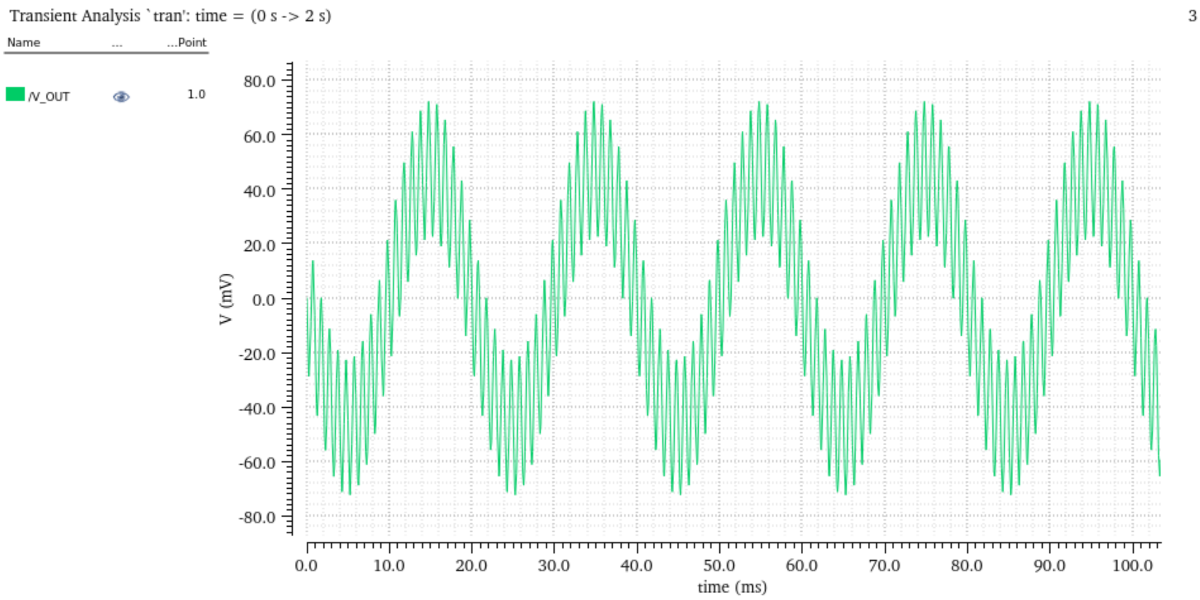


Figure 2.4: Plot of amplifier output showing combined effect of input and common-mode signals (showing degradation in CMRR).

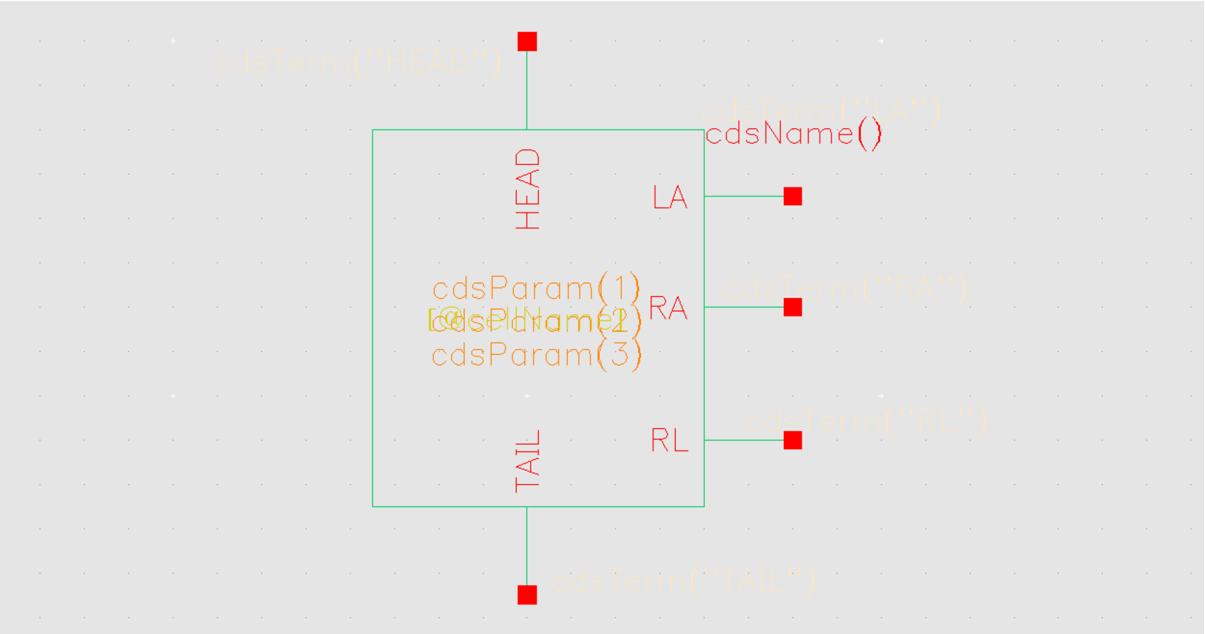
- Peak-to-Peak Value of Output (50 Hz): ______ V
- Simulated Gain at 50 Hz: ______

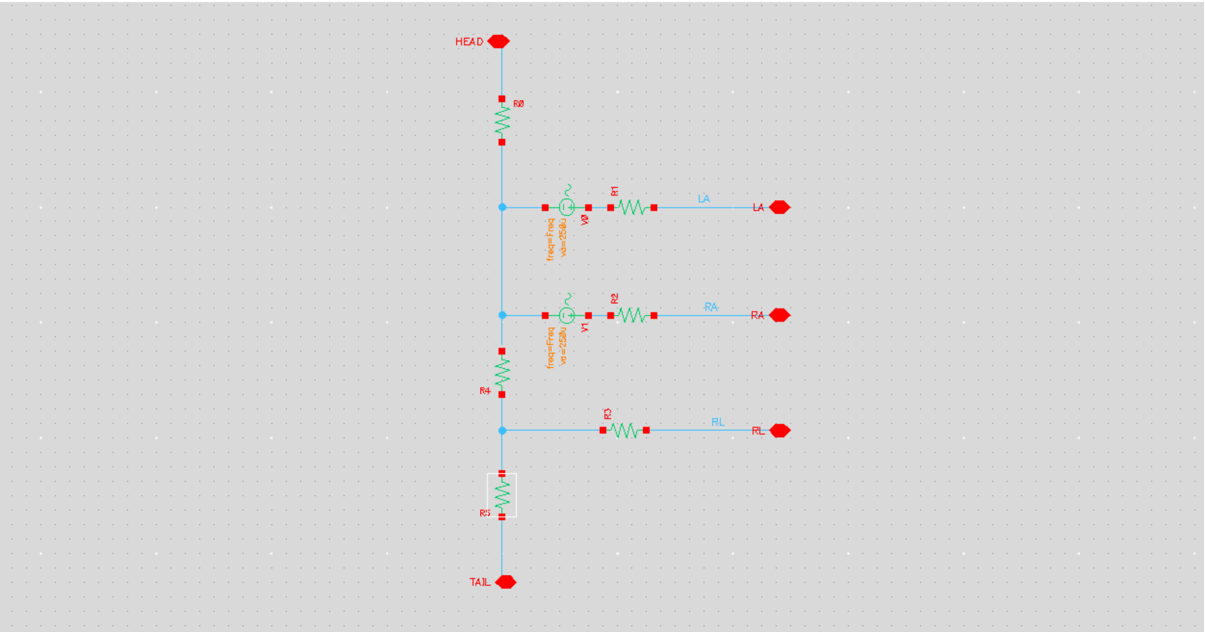
5. Discussion

Mismatch in R2 reduces CMRR, resulting in significant amplification of the common-mode component at 50 Hz.

Part 3: Body Model (ECG Source) Connection

1. Schematic of Body Model





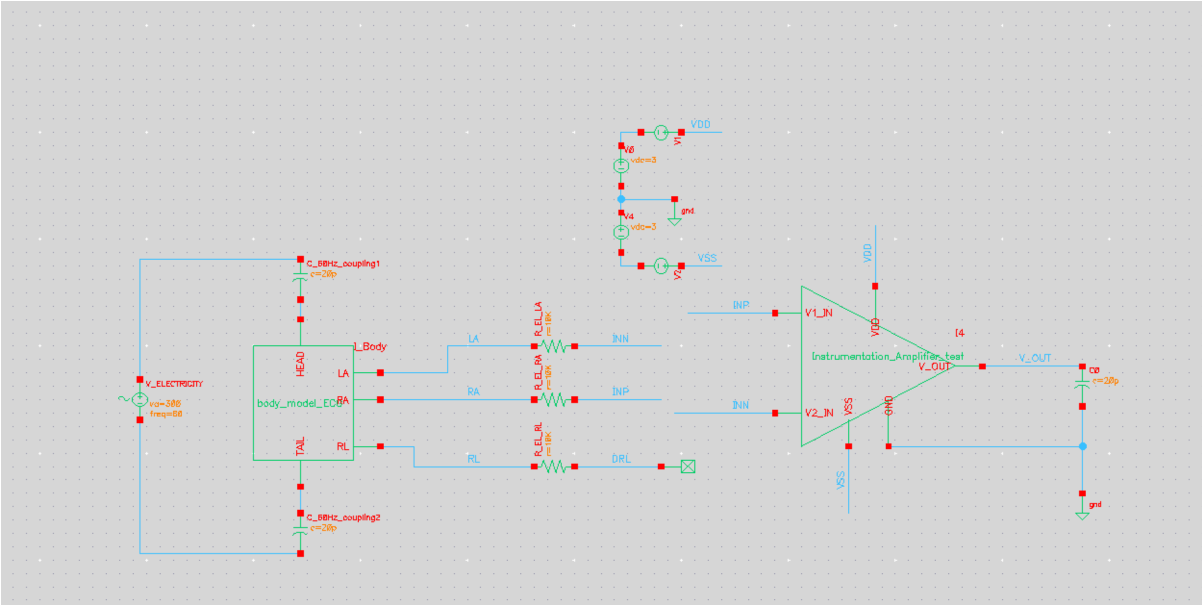
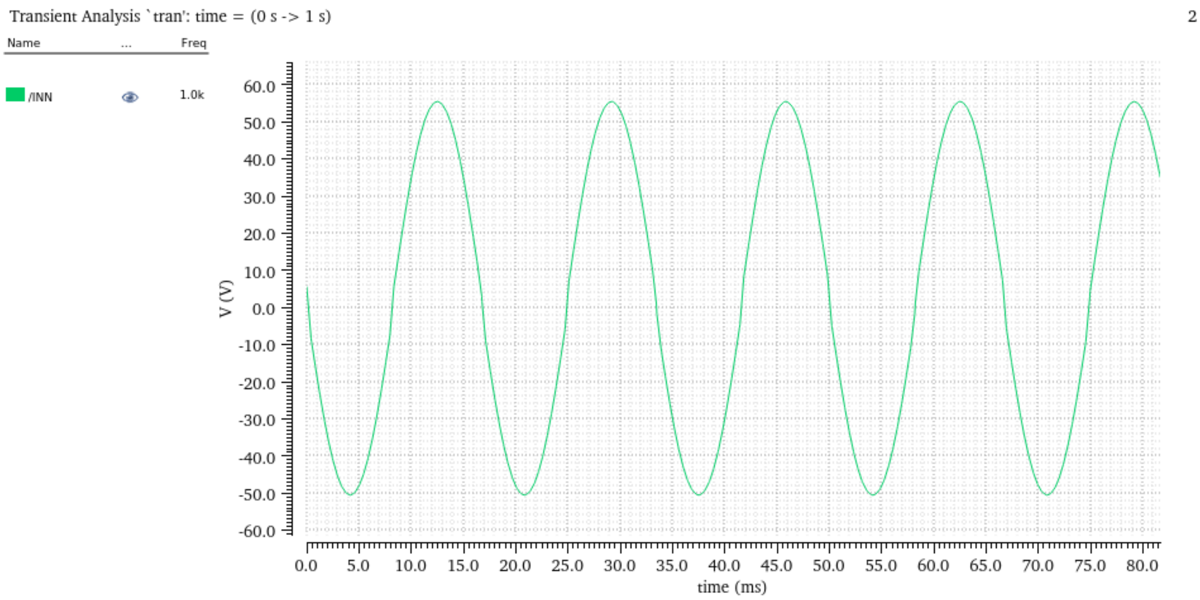


Figure 3.1: Schematic of Body_Model_ECG connected to IA with mismatch.

2. Input and Output Plots



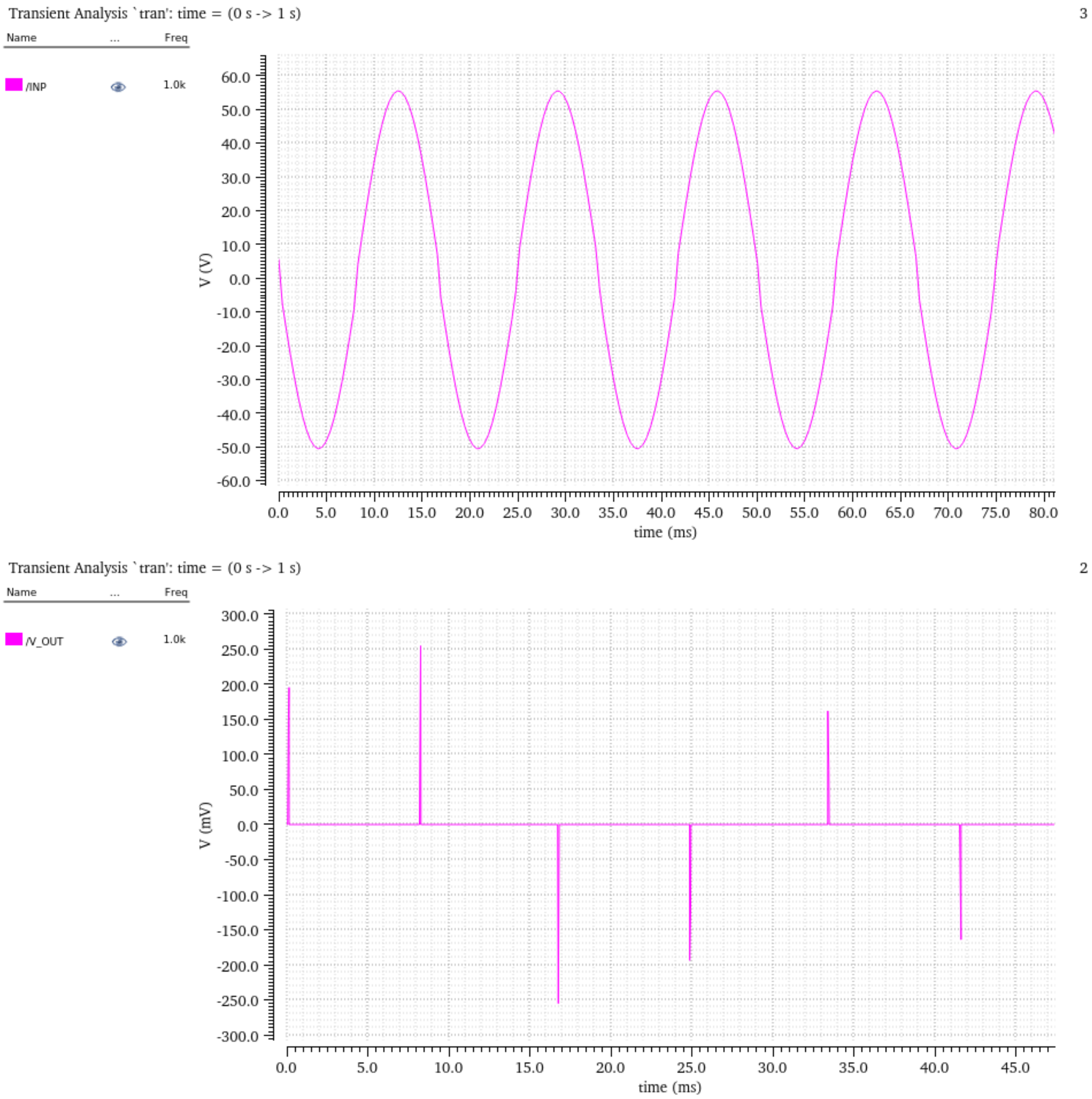


Figure 3.2: Plots of INP, INN, and IA output from Body_Model_ECG.

3. Discussion

The IA correctly amplifies the ECG differential input, but common-mode noise is visible due to mismatch.

Part 4: Body Model with DRL Node Grounded

1. Schematic of Body Model with DRL Grounded

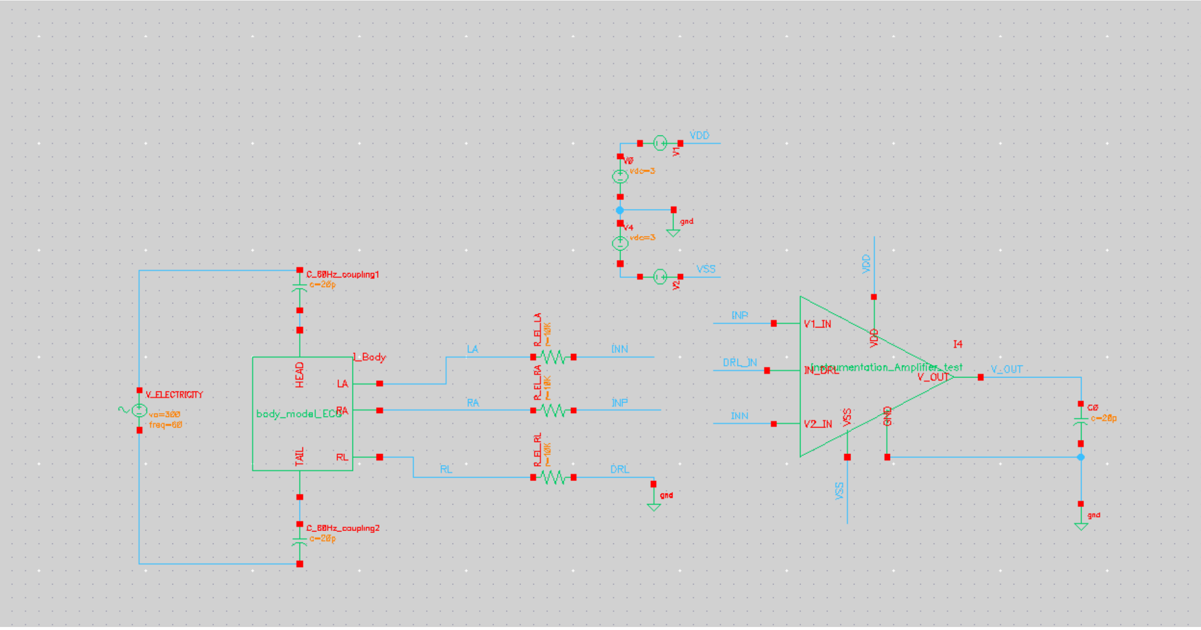
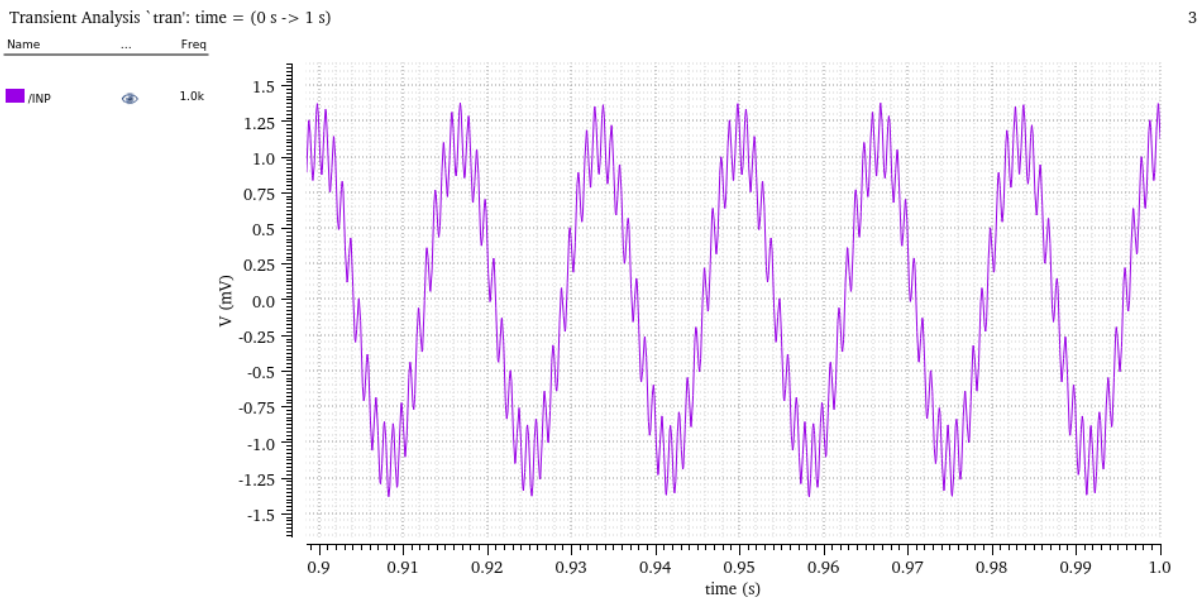


Figure 4.1: Test bench schematic with node DRL grounded.

2. Input and Output Plots



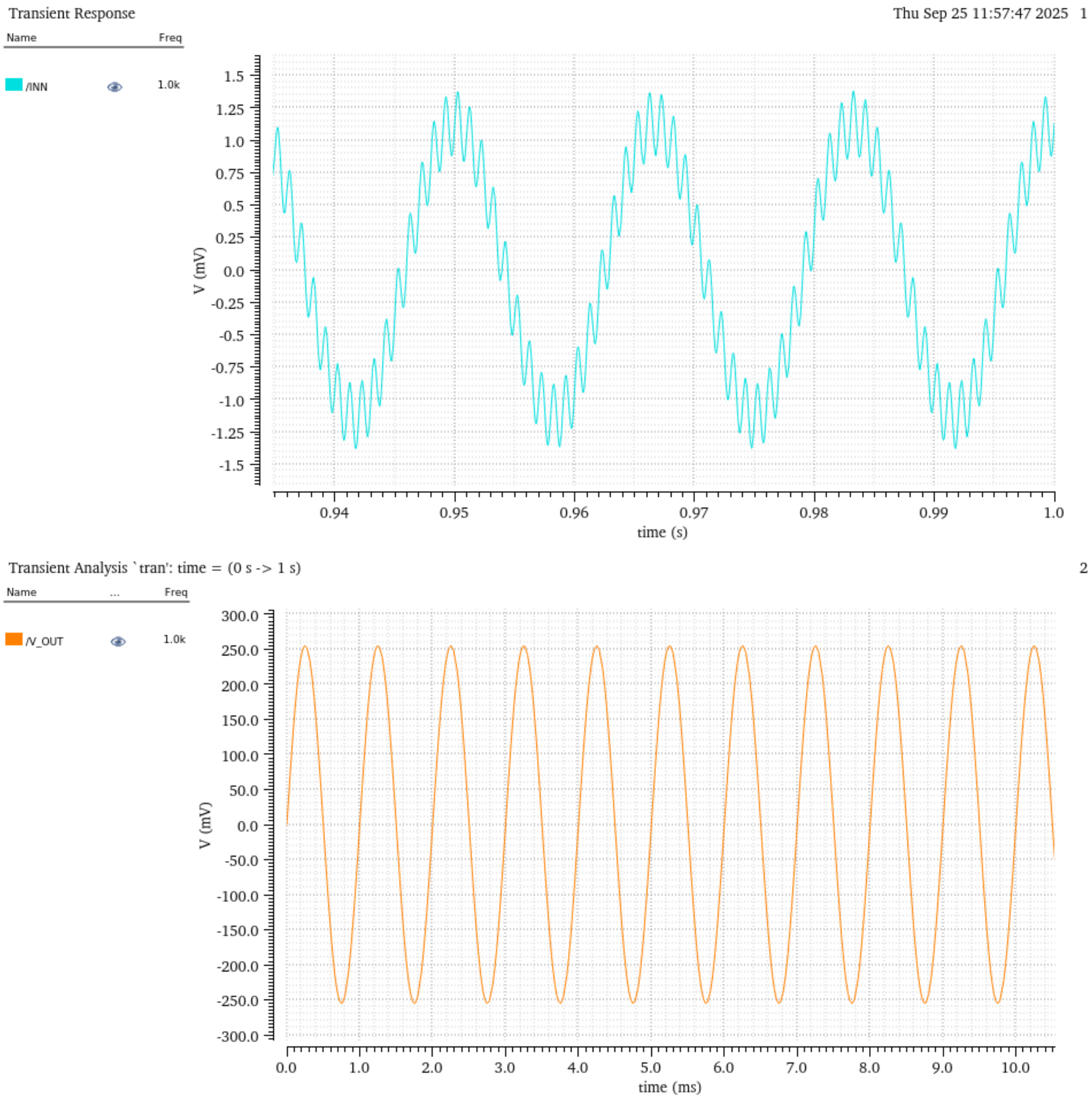


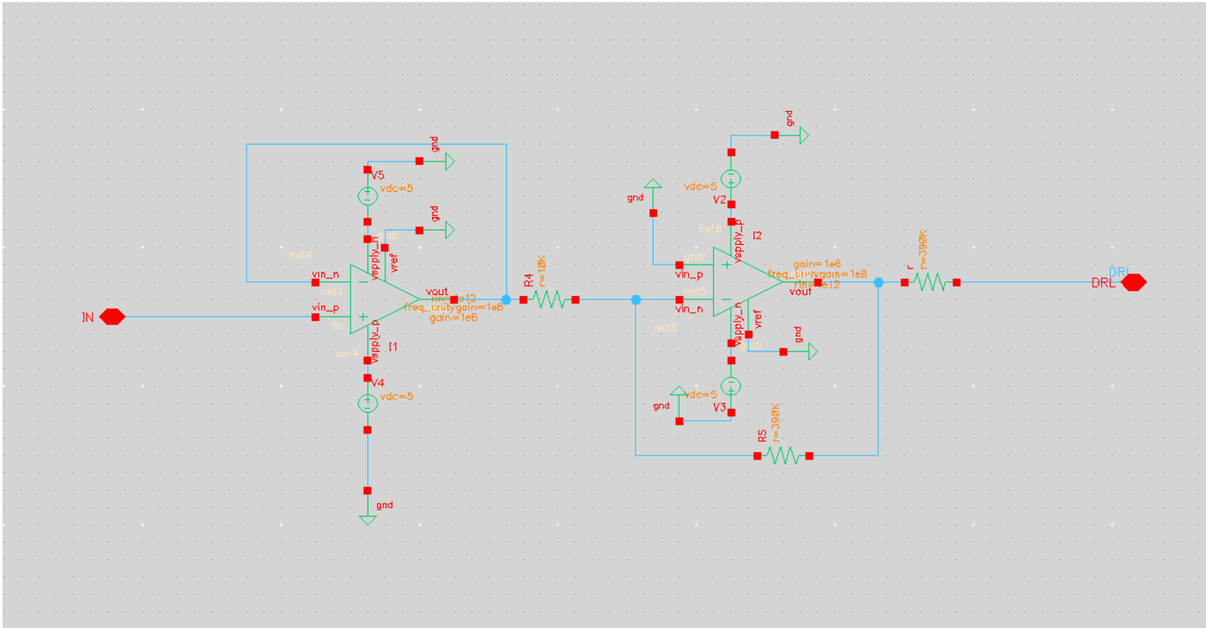
Figure 4.2: Plots of ECG inputs and output with DRL grounded.

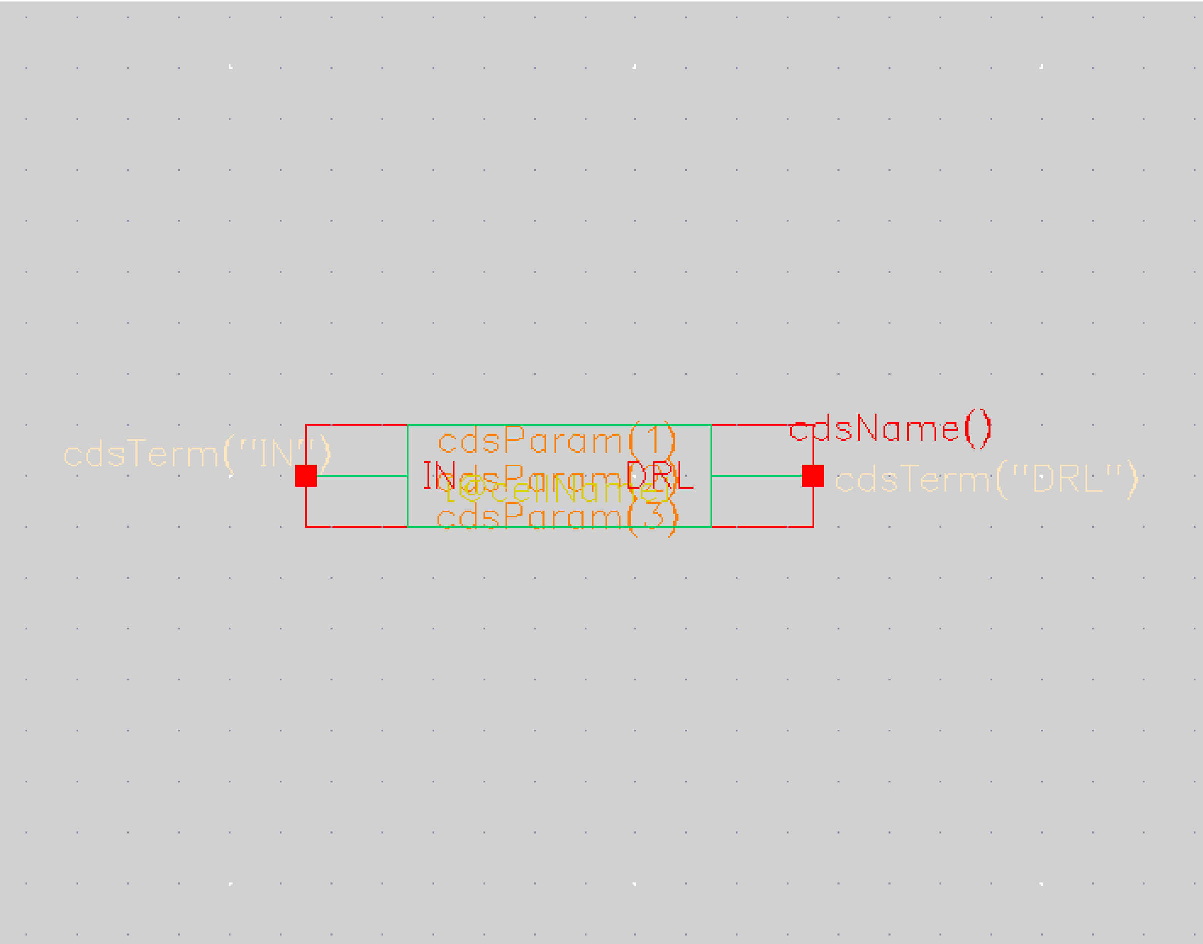
3. Discussion

Grounding DRL reduces stability and increases common-mode interference compared to the previous setup.

Part 6: Design of DRL Circuit

1. Schematic of DRL Circuit and Test Bench





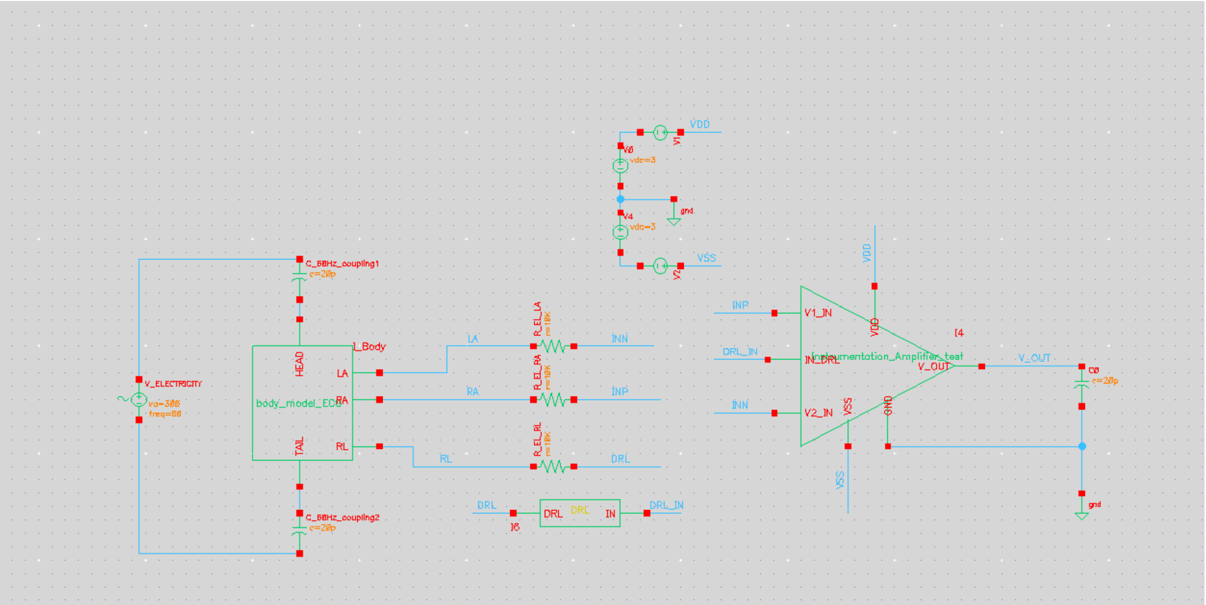
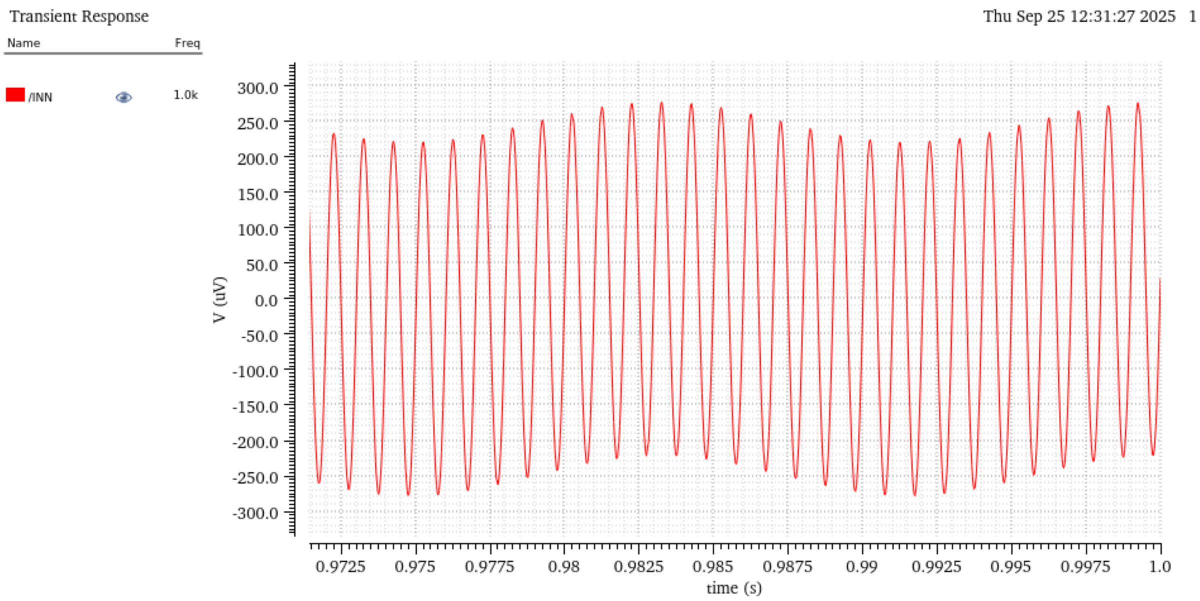
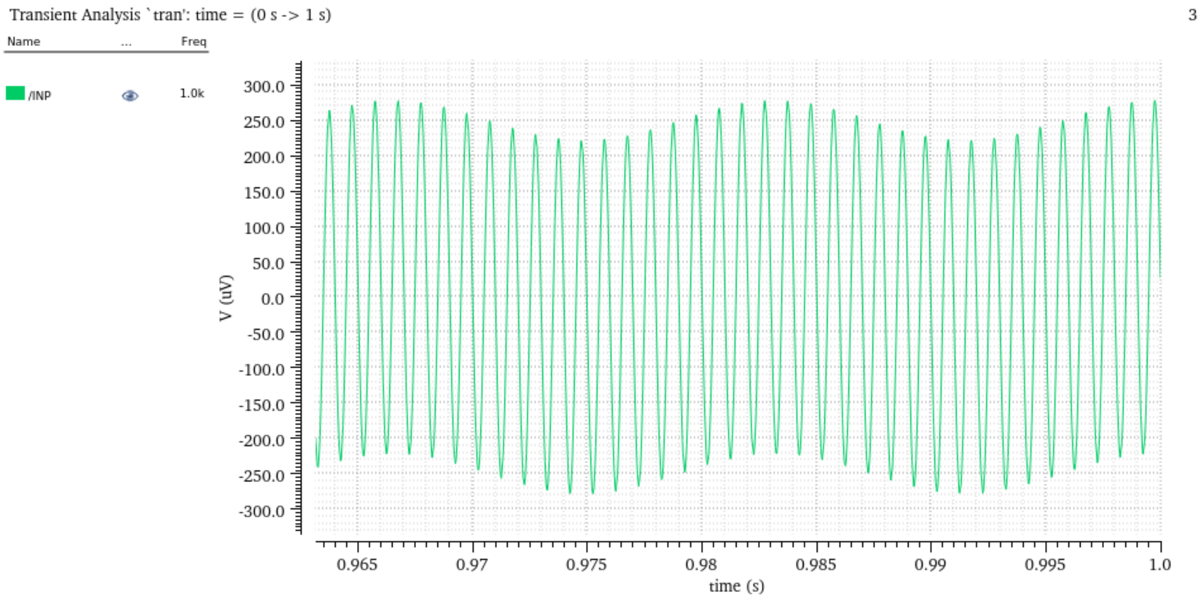


Figure 6.1: Test bench schematic with designed DRL feedback circuit.

2. Input and Output Plots





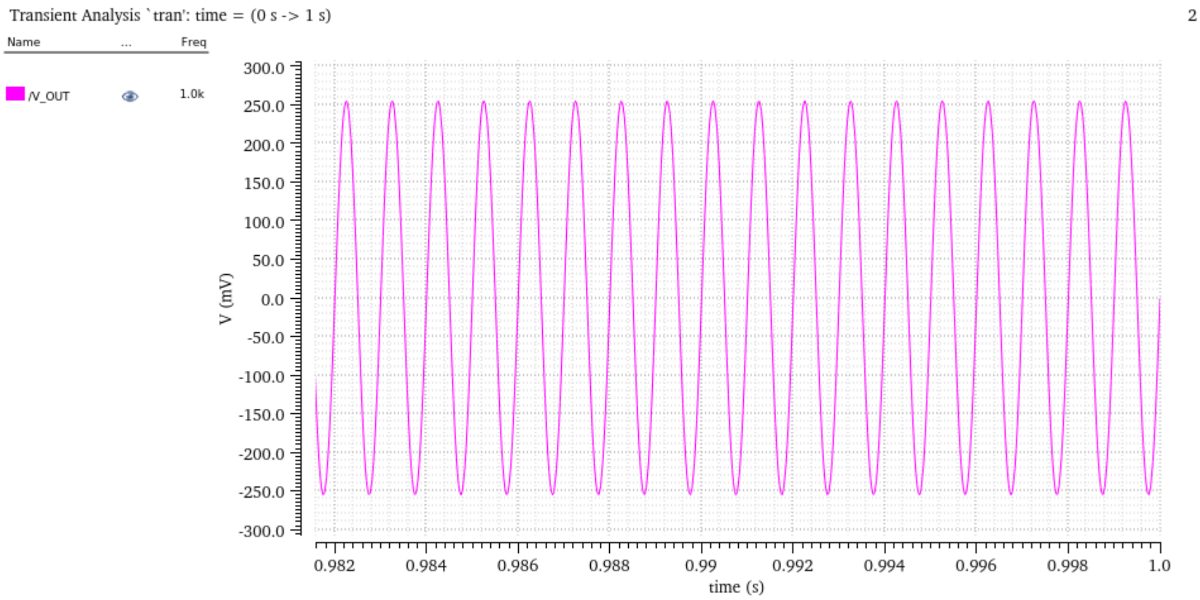


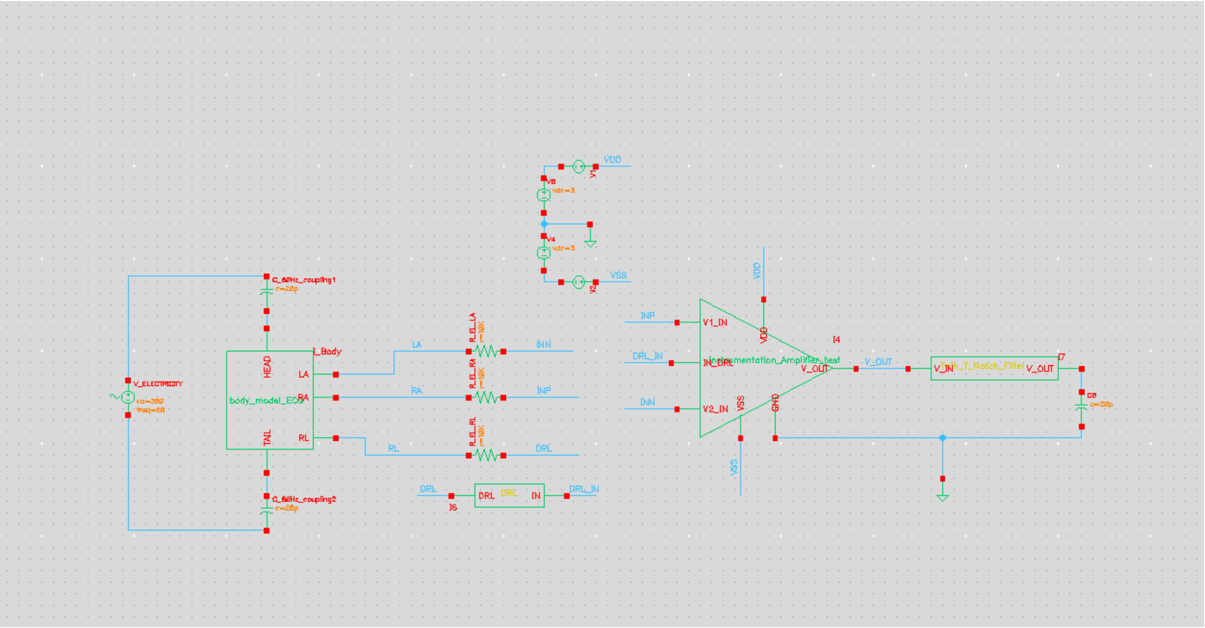
Figure 6.2: Plots of ECG inputs and output with DRL feedback applied.

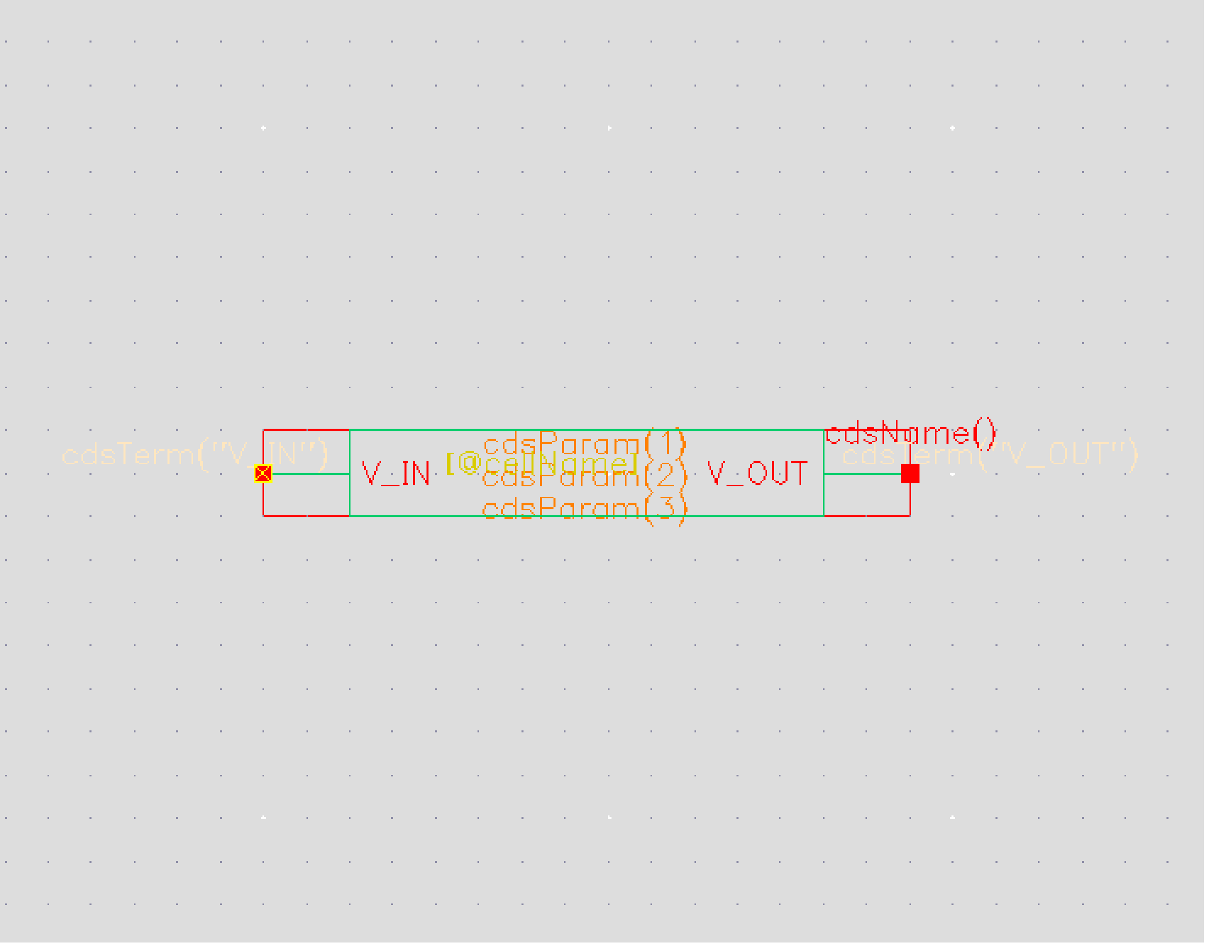
3. Discussion

DRL feedback significantly reduces common-mode interference compared to Part 3 and Part 4.

Part 7: Twin-T Notch Filter

1. Schematic of Twin-T Notch Filter with Buffer





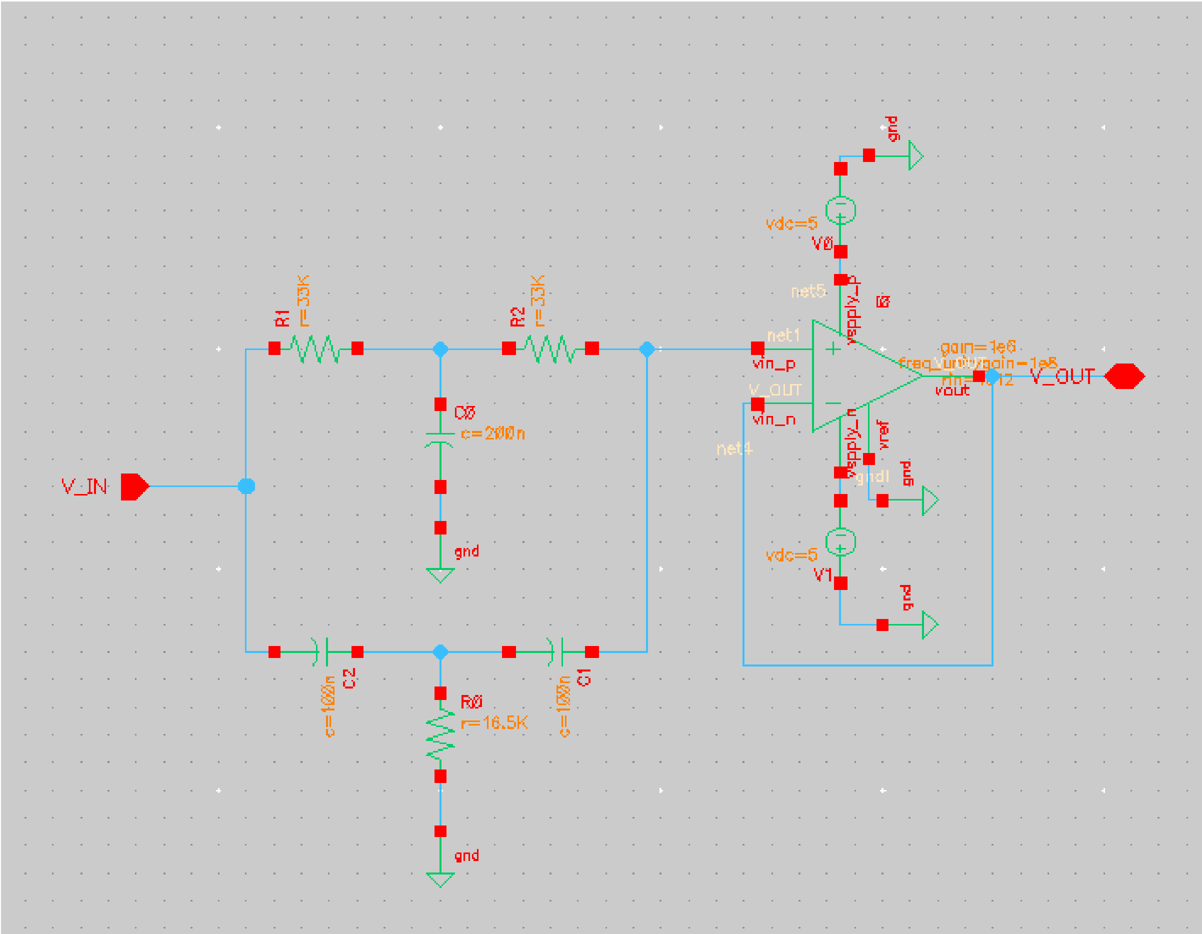
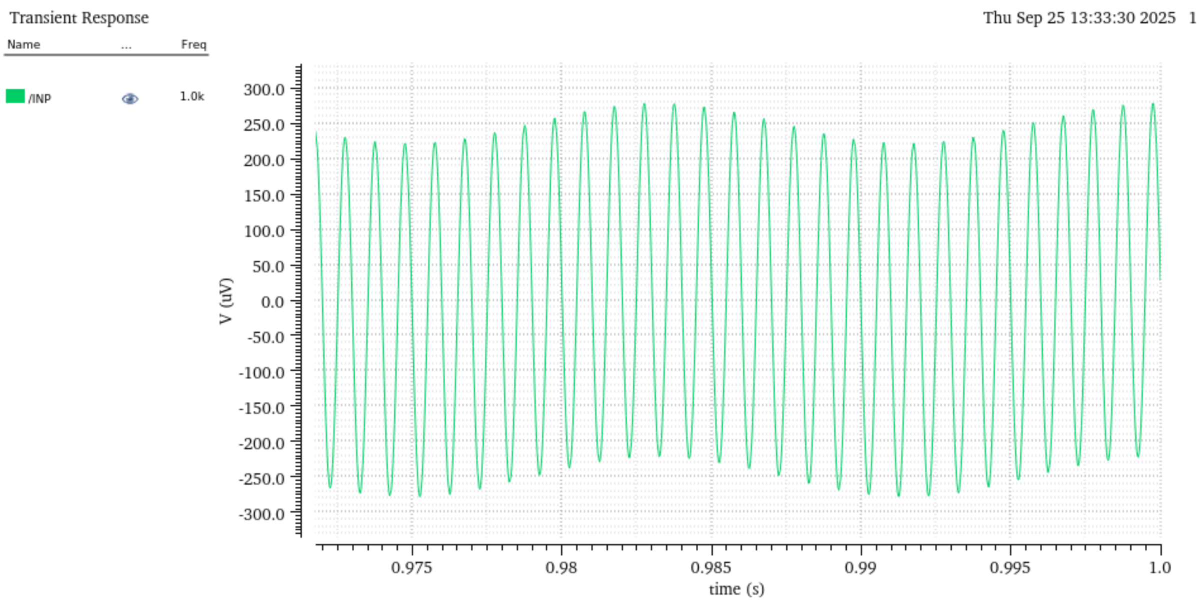


Figure 7.1: Test bench schematic of Twin-T notch filter with buffer stage.

2. Output Plot of Notch Filter



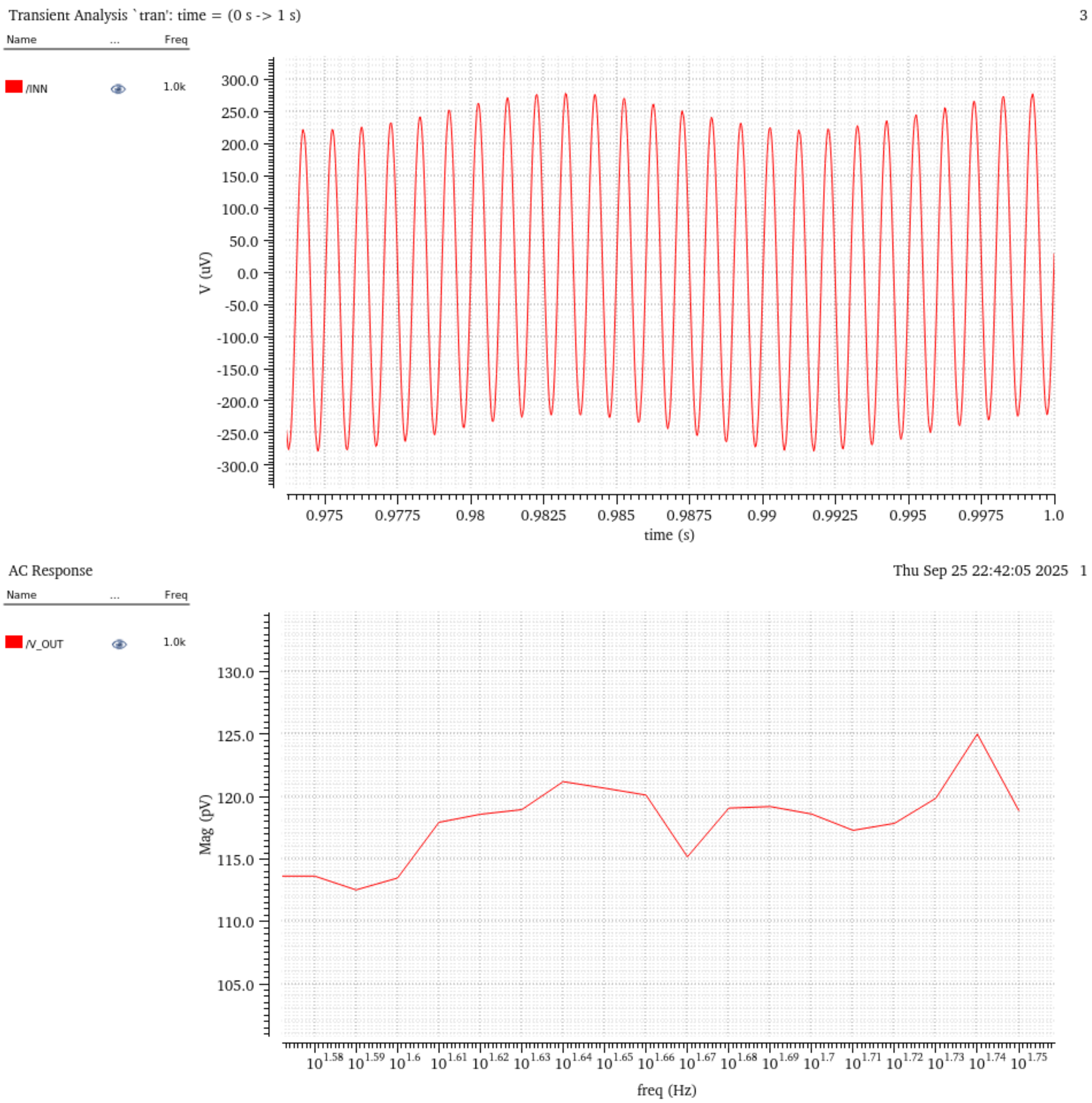


Figure 7.2: Output signal of notch filter showing suppression of 50 Hz component.

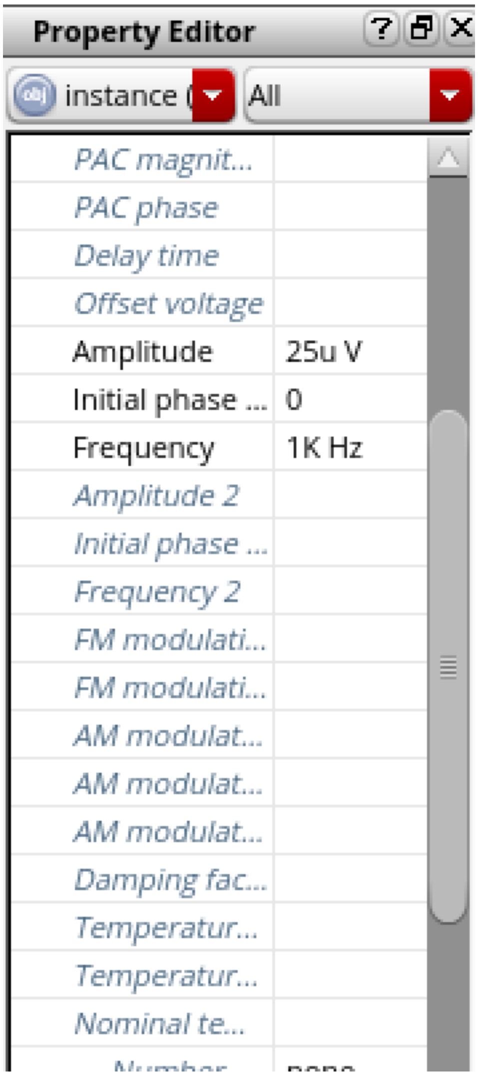
3. Discussion

The Twin-T notch filter effectively removes the 50 Hz noise, confirming its utility for ECG signal conditioning.

Conclusion

Across all simulations, the IA successfully amplified the small input signals while rejecting common-mode noise under ideal resistor matching. However, resistor mismatch reduced CMRR, making the system susceptible to power-line interference. The DRL circuit and

Twin-T notch filter provided significant improvement, demonstrating the importance of both active and passive filtering techniques in biomedical signal acquisition.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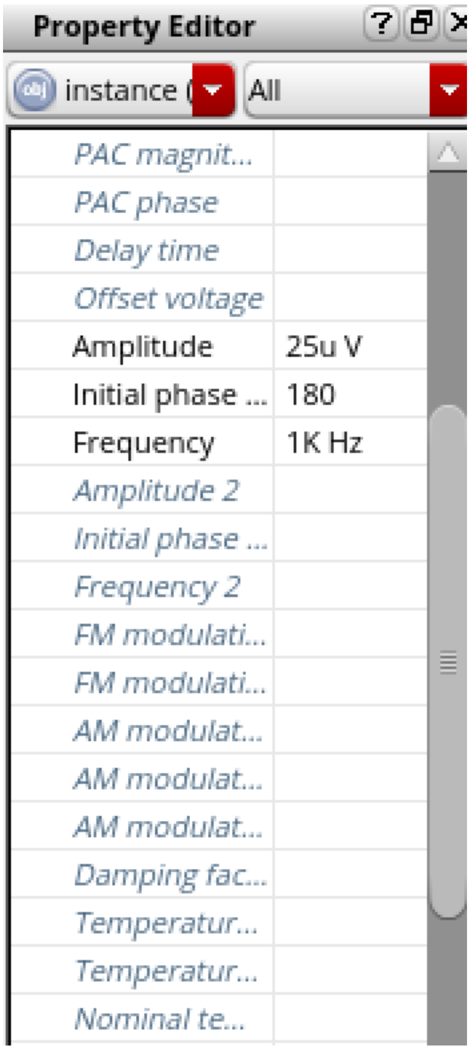
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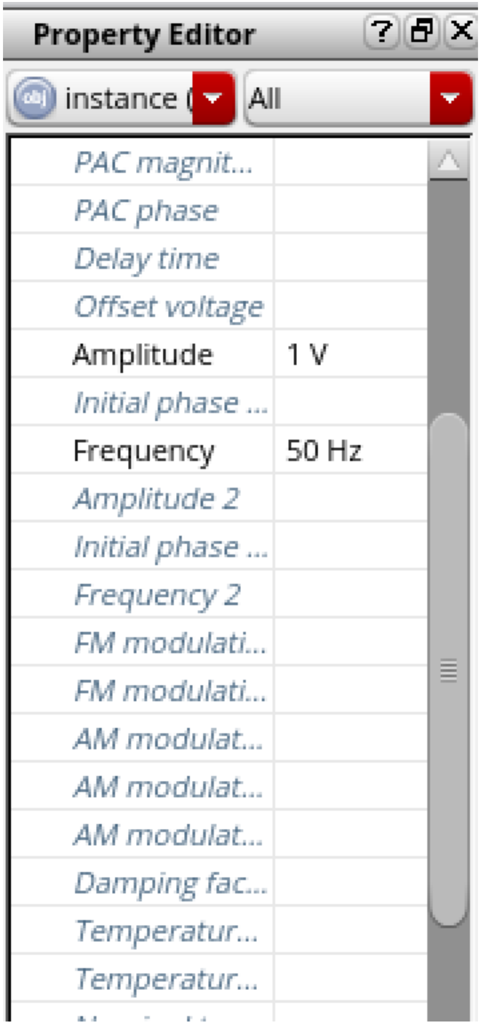
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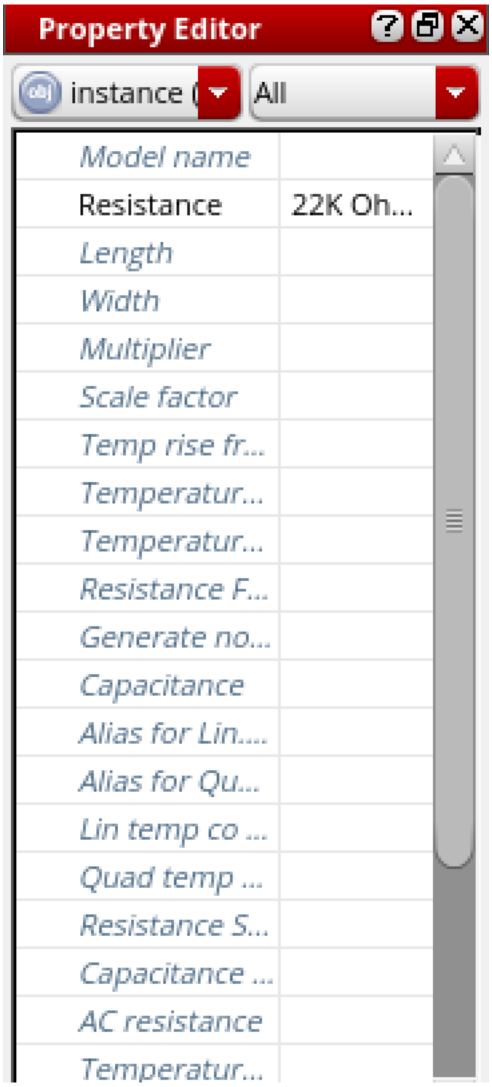
Part 2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AC resistance
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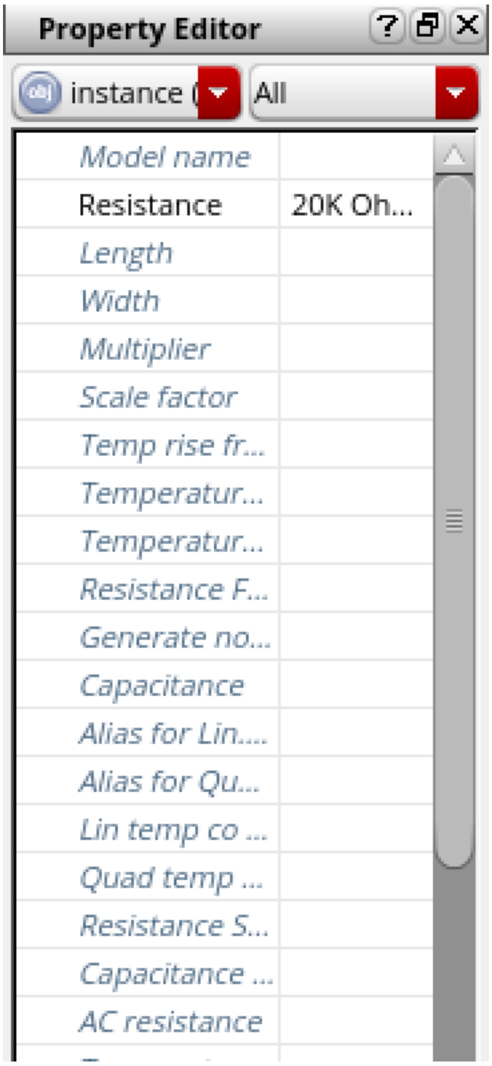
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Resistance S...
Capacitance ...
AC resistance

Part 3
Build the body model as shown below:

Now create a symbol for this cell as shown below and name it “Body_Model_ECG”.
Connect your IA as above with the mismatch, to the body model that you created. Please connect the two inputs of the IA to “*INP”* and “*INN”* in the “Body_Model_ECG ” schematic.
Run simulation.When you use the cell “body_model_ECG,” you need to set a parameter “Freq” as
follows:
What to present in your report:
1. 2. Schematics of the test bench.
Plots of the inputs and output separately.
3. Discuss the results.

Part 4
Now connect the node “DRL” to the ground. Run simulation.
What to present in your report:
1. 2. 3. Schematics of the test bench.
Plots of the inputs and output separately.
Discuss the results with comparison to the results of (e).

Part 6
Research and design a basic DRL circuit and connect its output to the node "*DRL*" in the schematic.
What to present in your report:
1. 2. 3. Schematics of the test bench.
Plots of the inputs and output separately.
Discuss the results with comparison to the results of (e)

Part 7

(15 pts) Build Twin –T Notch Filter with Buffer to filter out the 50 Hz frequency.
What to present in your report:
1. 2. Schematics of the test bench.
Run simulation and show plots of the output of the filter. Let us confirm that Notch filter was able to filter the 50 Hz noise introduced at the input.